

KAPADOKYA, Turkey

WMO: 171940

Lat: 38.767N

Lon: 34.533E

Elev: 945

StdP: 90.48

Time Zone: 3.00 (E03)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-9.2	-14.8	1.2	-9.8	-12.1	1.5	-6.3	12.4	4.5	10.1	5.2	2.5	120	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.1	36.0	17.1	34.2	16.7	32.8	16.5	18.9	30.8	18.2	29.7	17.5	28.9	4.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	12.1	21.5	14.2	11.3	21.0	13.6	10.9	20.9	57.5	30.9	55.2	29.5	53.0	28.9	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.7	7.4	DB	-15.5	38.3	3.8	1.5	-18.3	39.4	-20.5	40.3	-22.6	41.1	-25.4	42.2	(4)
(5)			WB	-15.9	20.4	3.7	1.5	-18.6	21.4	-20.7	22.3	-22.8	23.1	-25.5	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	0.9	3.6	7.7	11.9	16.7	21.0	23.9	24.3	19.5	14.0	7.4	3.0		
	DBStd	8.87	5.12	5.43	4.27	3.84	3.23	2.63	2.52	2.49	3.43	3.71	3.90	4.69		
	HDD10.0	908	282	183	96	23	1	0	0	0	0	10	95	220		
	HDD18.3	2529	540	413	331	197	72	6	0	0	27	139	327	477		
	CDD10.0	1954	0	3	24	78	207	329	431	443	286	134	18	2		
	CDD18.3	534	0	0	1	3	20	86	173	186	62	5	0	0		
	CDH23.3	7288	0	0	8	77	358	1134	2269	2414	905	123	0	0		
	CDH26.7	3282	0	0	1	14	94	442	1131	1233	347	19	0	0		
(14) Wind	WSAvg	3.1	3.2	3.0	3.5	3.3	3.1	3.4	3.7	3.5	2.9	2.7	2.6	2.6		
(15) Precipitation	PrecAvg	381	35	39	43	44	54	34	9	7	15	30	34	44		
	PrecMax	488	75	64	109	88	128	103	49	21	63	117	76	88		
	PrecMin	248	5	7	19	0	5	1	1	0	0	1	1	6		
	PrecStd	63	16	15	21	22	33	24	12	6	14	23	22	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	19.1	24.0	28.8	31.2	35.0	38.1	37.9	34.2	29.0	22.0	16.8		
		MCWB	6.7	8.6	11.1	12.4	14.9	16.1	17.4	17.6	16.0	13.5	11.0	9.1		
	2%	DB	11.9	16.2	20.2	25.2	29.1	33.0	36.1	36.1	32.2	26.8	19.2	13.2		
		MCWB	6.2	7.8	9.1	12.0	14.3	16.3	17.1	17.2	15.3	13.2	10.0	7.6		
	5%	DB	9.8	14.0	18.0	23.0	27.2	31.1	34.1	34.8	30.2	24.2	17.1	11.2		
		MCWB	5.6	7.2	8.6	11.2	13.9	15.8	16.9	16.9	14.9	12.9	8.9	6.8		
	10%	DB	7.8	11.8	15.9	20.8	25.1	29.2	32.2	33.0	28.2	22.2	14.9	9.2		
		MCWB	4.6	6.4	7.8	10.5	13.6	15.3	16.5	16.8	14.4	12.5	8.6	5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	10.1	11.2	13.4	16.6	18.8	19.7	19.9	17.8	15.5	12.3	9.6		
		MCDB	12.8	16.6	21.5	24.2	25.9	29.0	31.7	31.9	29.6	22.3	19.8	14.6		
	2%	WB	6.7	8.4	10.1	12.6	15.6	17.9	18.8	19.0	16.5	14.4	10.9	8.4		
		MCDB	10.5	14.7	18.1	23.7	25.2	27.7	31.2	31.0	27.9	22.3	17.1	12.5		
	5%	WB	5.7	7.5	9.1	11.7	14.7	17.1	18.0	18.3	15.8	13.7	9.8	7.4		
		MCDB	8.7	13.2	16.4	21.4	24.2	26.4	29.8	30.1	27.3	21.8	15.5	10.8		
	10%	WB	4.7	6.4	8.3	10.9	14.1	16.4	17.3	17.6	15.1	13.1	8.9	6.2		
		MCDB	7.3	10.9	14.9	19.2	23.2	25.8	29.1	29.4	26.2	20.7	14.2	8.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	10.9	12.0	13.6	14.3	14.9	15.7	16.1	15.9	14.2	12.8	9.3		
		MCDBR	11.3	14.1	15.8	17.8	17.6	17.6	18.3	18.3	18.6	17.2	16.1	12.1		
	5% WB	MCWBR	6.9	7.3	7.4	7.4	6.3	5.6	5.7	6.3	7.3	7.9	8.4	7.2		
		MCWBR	9.7	12.7	13.8	15.8	15.5	15.0	15.9	15.6	15.8	14.6	13.8	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.331	0.370	0.411	0.389	0.360	0.359	0.360	0.337	0.347	0.315	0.299		
		taud	2.393	2.321	2.227	2.135	2.266	2.364	2.365	2.330	2.400	2.368	2.427	2.445		
	Ebn at Noon		872	888	883	865	891	914	913	904	910	859	850	850		
		Edh at Noon	86	106	129	150	135	122	121	122	107	99	82	76		
(44) All-Sky Solar Radiation	RadAvg		2.12	3.12	4.20	5.31	6.33	7.34	7.68	6.94	5.53	3.80	2.62	1.90		
	RadStd		0.25	0.41	0.35	0.45	0.37	0.49	0.32	0.20	0.32	0.23	0.25	0.23		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page